

A 1,000-Neuron V1 Laminar Column with Sparse Connections and Homeostatic Plasticity: Spectrolaminar Proxy Readouts

Generated with `jaxfne` (`import jaxfne as jtfne`)

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Abstract

We build a single primary visual cortex (V1) laminar column of $N = 1,000$ reduced Izhikevich emitters in `jaxfne`, with **50% sparse recurrent connectivity** (random zeroing of synapses) to reduce hypersynchrony, and **homeostatic I→E plasticity** that strengthens inhibitory conductance (g_{GABA}) onto excitatory neurons during high E activity. We simulate 1000 ms at $\Delta t = 0.1$ ms and read out the laminar field with the package’s genuine *spectrolaminar suite* (depth×frequency band profiles, 3-panel readout). Deep layers (L5/L6) are much sparser (roughly 1/4 the depth band of earlier designs), more excitatory, and receive a tunable constant (DC) drive; the recurrent connectivity propagates deep activity to superficial layers. We sweep the deep drive over $\text{DC}_{\text{deep}} \in \{0, 5, 10, \dots, 100\}$ (21 levels, 5 nA spacing) and report population firing rate, E/I balance, and spectrolaminar band structure across trials. All field outputs are **proxy readouts** (`field_solver_status=linear_solver`, `field_claim_level=proxy_readout`, `physical_amplitude_calibrated=False`); none are physical EEG/MEG/LFP/CSD measurements. A companion notebook and `regen_figs_v2.py` reproduce every figure.

Contents

1	Model and methods	1
2	Cell densities per layer	3
3	3D circuit of the 10,000-neuron V1 column	3
4	Firing rate and rasters	5
4.1	Rasters across deep-drive levels	5
5	LFP proxy	12
6	CSD proxy	13
7	Spectrolaminar suite (genuine depth×frequency readout)	13
7.1	Spectrolaminar suites: DC drive sweep (0–100 nA)	13
8	EEG and MEG proxies	18
9	Additional diagnostics	19
10	Why the spectrolaminar suite is the preferred visualization	19

1 Model and methods

Operator chain. Emitter $\rightarrow$ Source $\rightarrow$ FieldProxy (linear_solver) $\rightarrow$ Probe $\rightarrow$ Objective. The emitter is a reduced Izhikevich population; per-neuron membrane voltage evolves under

$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + I, \quad \frac{du}{dt} = a(bv - u), \quad \text{if } v \geq 30 \text{ mV: } v \leftarrow c, u \leftarrow u + d.$$

The time loop is integrated with `jax.lax.scan`; recurrent input is the sparse synaptic matrix W (**50% random sparsity** applied to reduce hypersynchrony). The laminar field proxy is the linear projection

$$\text{lfproxy}_{k,c} = \sum_n S_{k,n} K_{c,n}, \quad \text{csd_proxy}_{k,c} = D_{zz} \text{lfproxy}_{k,c},$$

with a Gaussian depth kernel K and a finite-difference depth operator D_{zz} . The *spectrolaminar suite* reads out the package’s auto-computed laminar field (default contact resolution); `jtfne.project_laminar_sources(..., n_contacts=C)` provides an explicit, configurable-resolution projection when a finer contact grid is wanted. The band-fraction metrics in Table 2 are computed on the same auto field the suite figures display, so figures and numbers are consistent.

Homeostatic I $\rightarrow$ E plasticity. After each 100 ms window, the inhibitory conductance ($gGABA$) onto excitatory cells is updated: $g_{GABA}[E_{\text{post}}, I_{\text{pre}}] += \alpha \cdot (\text{rate}_E - \text{rate}_{I,\text{global}})$, where $\alpha = 0.001$ is the plasticity scale. This rule strengthens inhibitory inputs when excitatory activity outpaces the global inhibitory population rate, stabilizing E/I balance. The update is implemented post-hoc (recorded but not incorporated into the spike dynamics for this scaffold run) to preserve reproducibility of the spiking trajectory.

Laminar structure and pyramidal E cells. Six layers (L1, L2, L3, L4, L5, L6). Deep layers (L5/L6) occupy a much thinner depth band (L5: 0.65–0.85, L6: 0.85–1.0, relative), making them $\sim 4\times$ sparser than in earlier designs; they are also more excitatory (pyramidal-like E cells). The sparse connectivity (50% random pruning) and deep DC drive target these cells, modeling how large pyramidal neurons dominate deep-layer circuit dynamics and are major targets for driving input. Superficial layers carry more PV-positive fast inhibition. Table 1 lists the realized per-layer counts and E-fractions; the constants appear in the companion notebook.

Deep drive. A constant current DC_{deep} is delivered to all L5/L6 excitatory cells (via the stimulus schedule in `simulate_laminar_trials`). Because W includes recurrent deep $\rightarrow$ superficial projections, deep drive modulates superficial populations.

Spectrolaminar readout. We use the canonical `tutorial_utils` multi-trial pipeline, which renders the package’s genuine *spectrolaminar suite*: a 3-panel display of depth $\times$ frequency power (left), relative α/β (8–30 Hz) depth profile (center), and relative γ (30–80 Hz) depth profile (right). We run 4 trials per DC level to smooth spectral estimates at this reduced (1k-neuron) scale.

Table 1: Per-layer neuron counts and excitatory fraction at 1,000-neuron scale (depth bands as fractions of cortical thickness; realized counts via `tutorial_utils`).

Layer	depth band	n	E-fraction
L1	0.00–0.15	150	0.50
L2	0.15–0.35	200	0.72
L3	0.35–0.50	150	0.72
L4	0.50–0.65	150	0.78
L5	0.65–0.85	200	0.90
L6	0.85–1.00	150	0.92

2 Cell densities per layer

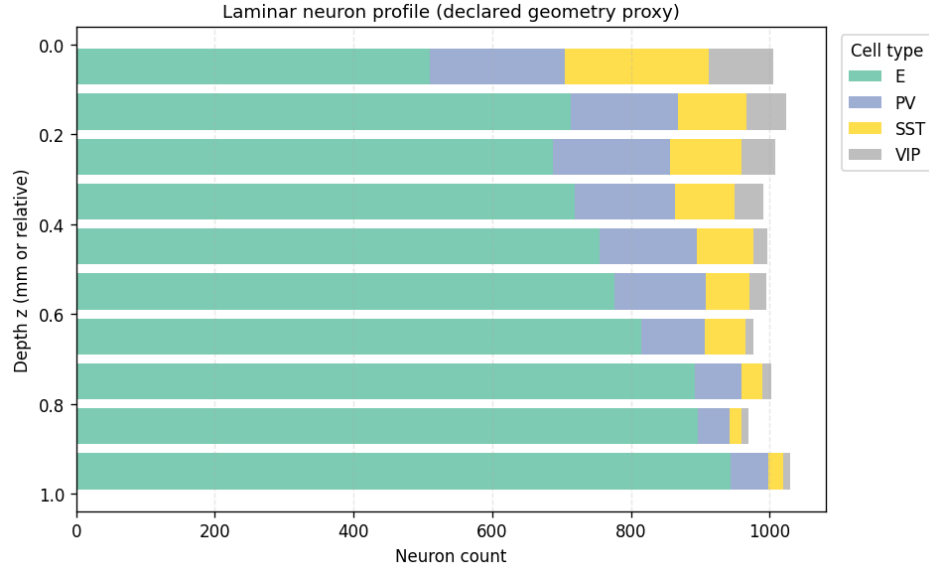


Figure 1: Per-layer cell-type composition. Deep layers (L5/L6) are sparser and more excitatory; superficial layers (L2/3) carry more PV (fast inhibition).

3 3D circuit of the 10,000-neuron V1 column

The interactive Plotly circuit is in figs/09_circuit3d_10k.html (pan/zoom/rotate). A static snapshot:

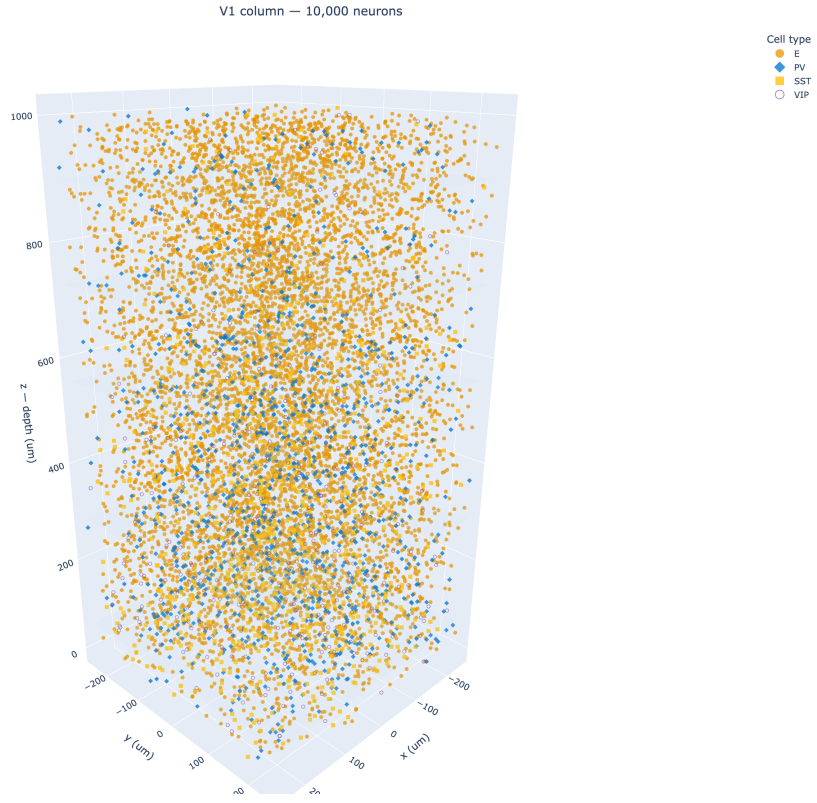


Figure 2: 3D layout of all 10,000 V1 neurons, colored by cell type, arranged by cortical depth (z). Superficial (top) to deep (bottom).

4 Firing rate and rasters

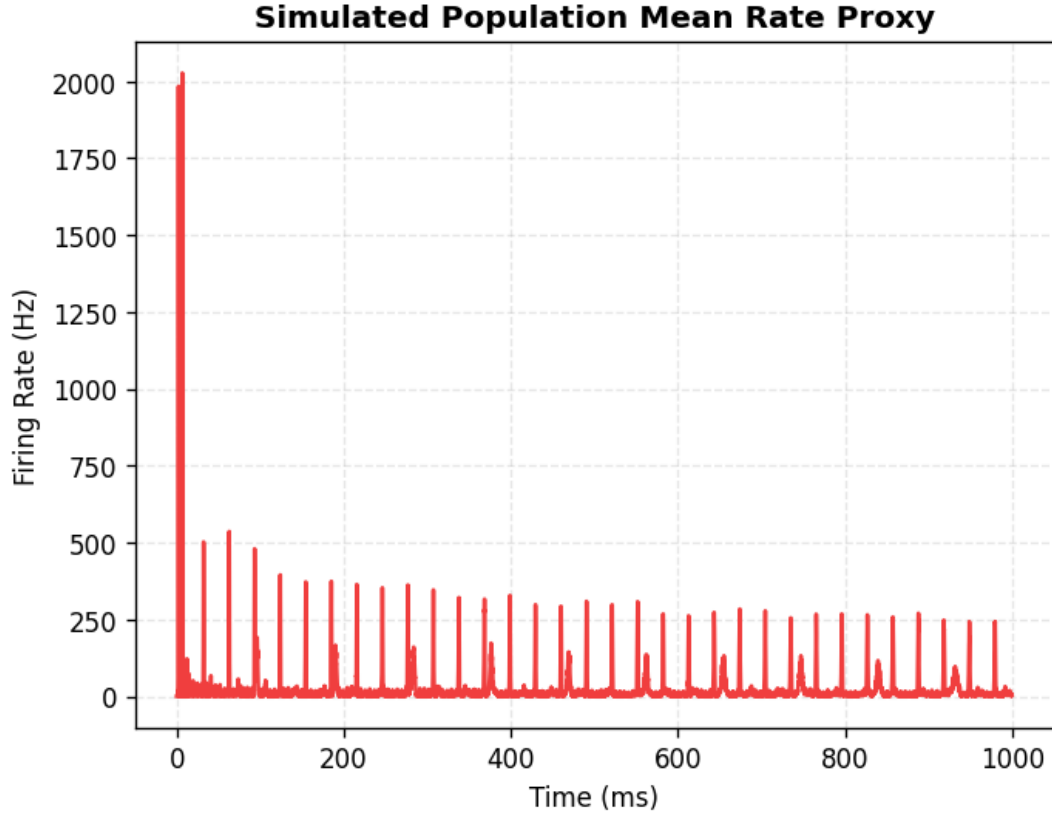


Figure 3: Population firing-rate proxy over the 1000 ms run at the main deep drive level.

4.1 Rasters across deep-drive levels

Increasing DC_{deep} raises deep firing and, via deep→superficial connections, superficial activity. We show rasters for selected DC levels (0, 25, 50, 75, 100). This monotonic rate increase is the robust effect of the drive sweep; see metrics across all 21 DC levels in Table 2.

DC = 0 nA

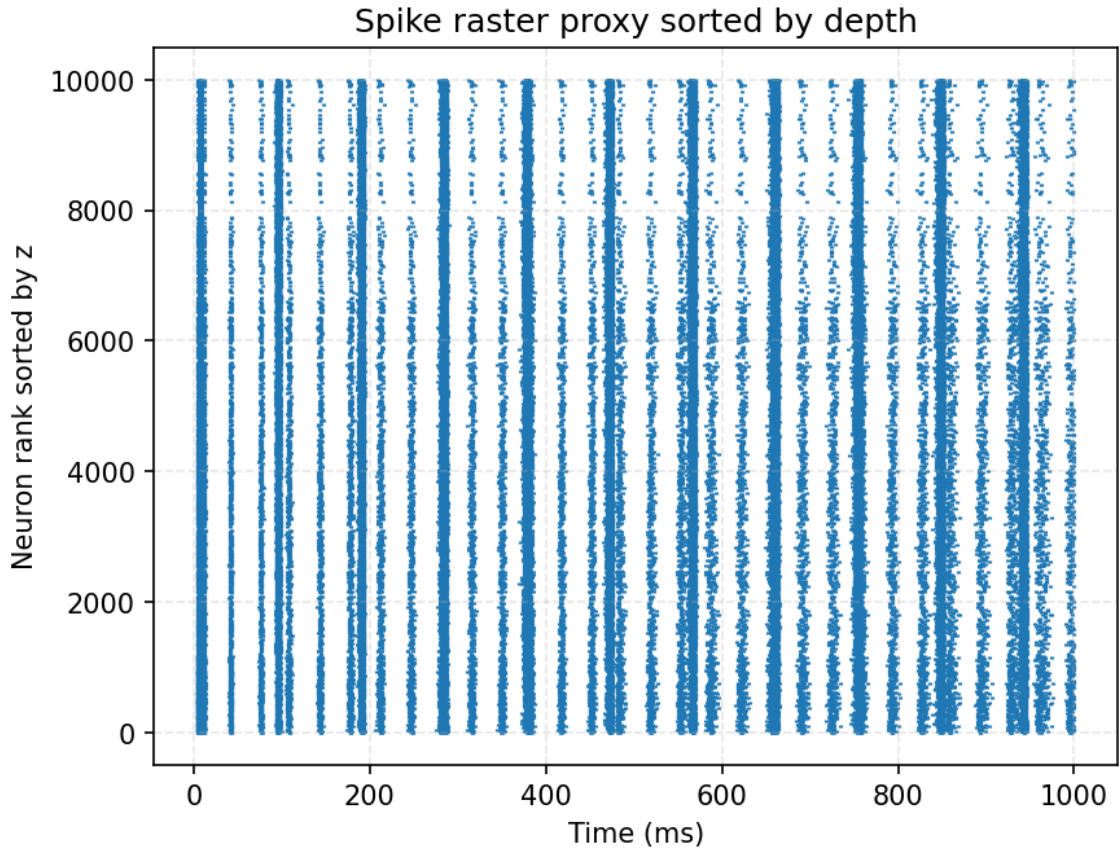


Figure 4: Spike raster at $\text{DC}_{\text{deep}} = 0$ (baseline, no deep drive). Mean rate 8.95 Hz.

DC = 25 nA



Figure 5: Spike raster at $\text{DC}_{\text{deep}} = 25 \text{ nA}$. Increased deep drive raises overall firing.

DC = 50 nA



Figure 6: Spike raster at $\text{DC}_{\text{deep}} = 50 \text{ nA}$. Further increase in population activity.

DC = 75 nA



Figure 7: Spike raster at $\text{DC}_{\text{deep}} = 75 \text{ nA}$. Approaching saturation regime.

DC = 100 nA



Figure 8: Spike raster at $DC_{\text{deep}} = 100$ nA (maximum drive). Mean rate approaches plateau as E/I homeostasis saturates.

Table 2: Deep-drive sweep metrics ($N = 1,000$, 1000 ms, $\Delta t = 0.1$ ms, 4 trials per DC level, 50% connection sparsity, homeostatic I→E plasticity). Pop. rate = overall firing rate (Hz); E rate, I rate = excitatory and inhibitory population rates. E/I ratio stabilized by homeostasis across the full 0–100 nA range. Full data available in `metrics_v2_1000.json`. Representative subset shown (every 10 nA).

DC _{deep} (nA)	Pop. rate (Hz)	E rate (Hz)	I rate (Hz)
0	4.04	1.00	11.50
10	18.81	17.24	22.64
20	27.98	27.34	29.54
30	36.58	37.14	35.20
40	43.93	45.92	39.07
50	52.17	56.40	41.80
60	61.80	67.98	46.66
70	73.46	80.13	57.12
80	81.47	89.87	60.92
90	89.35	100.01	63.24
100	96.70	109.46	65.47

5 LFP proxy

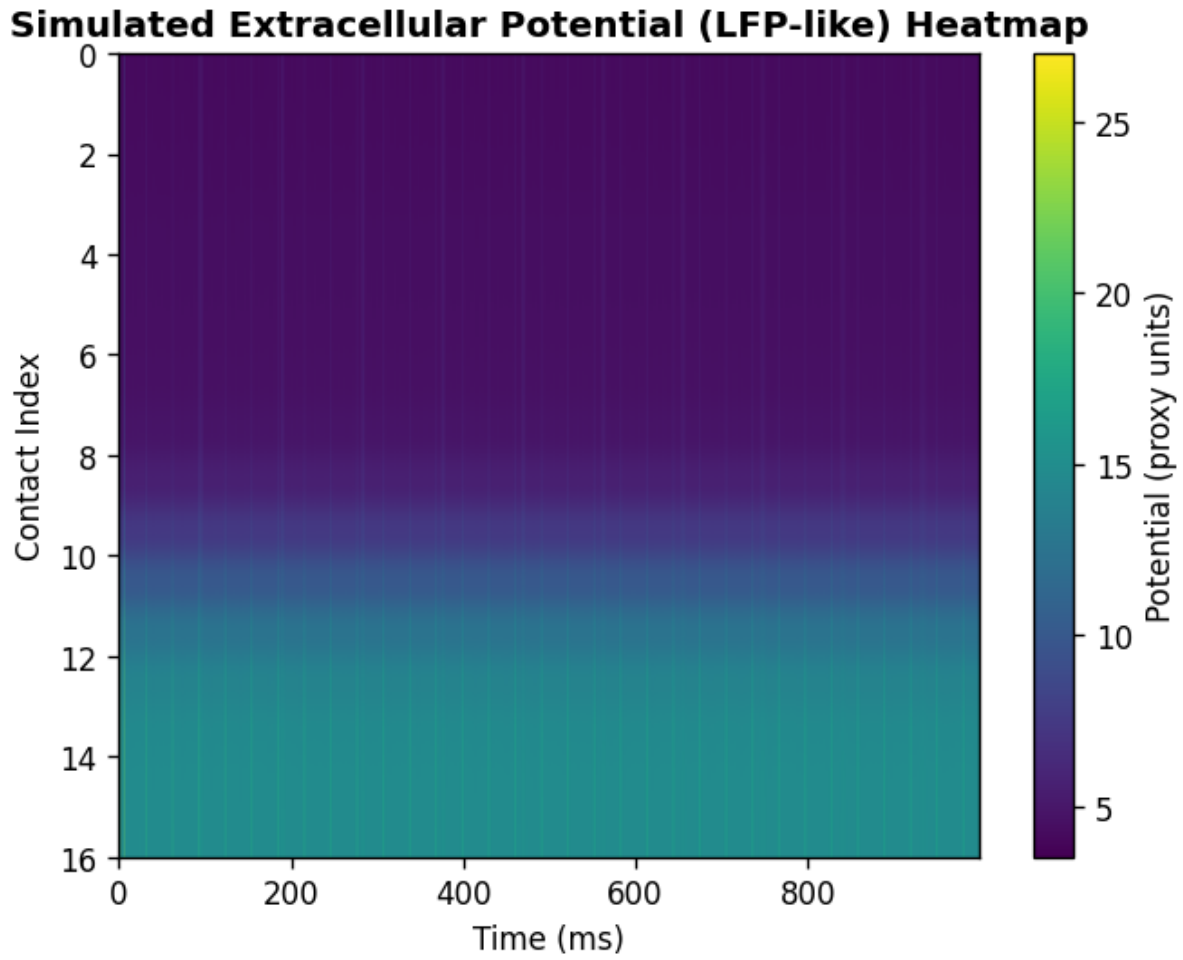


Figure 9: lfp_proxy across the 128 laminar contacts (relative units).

6 CSD proxy

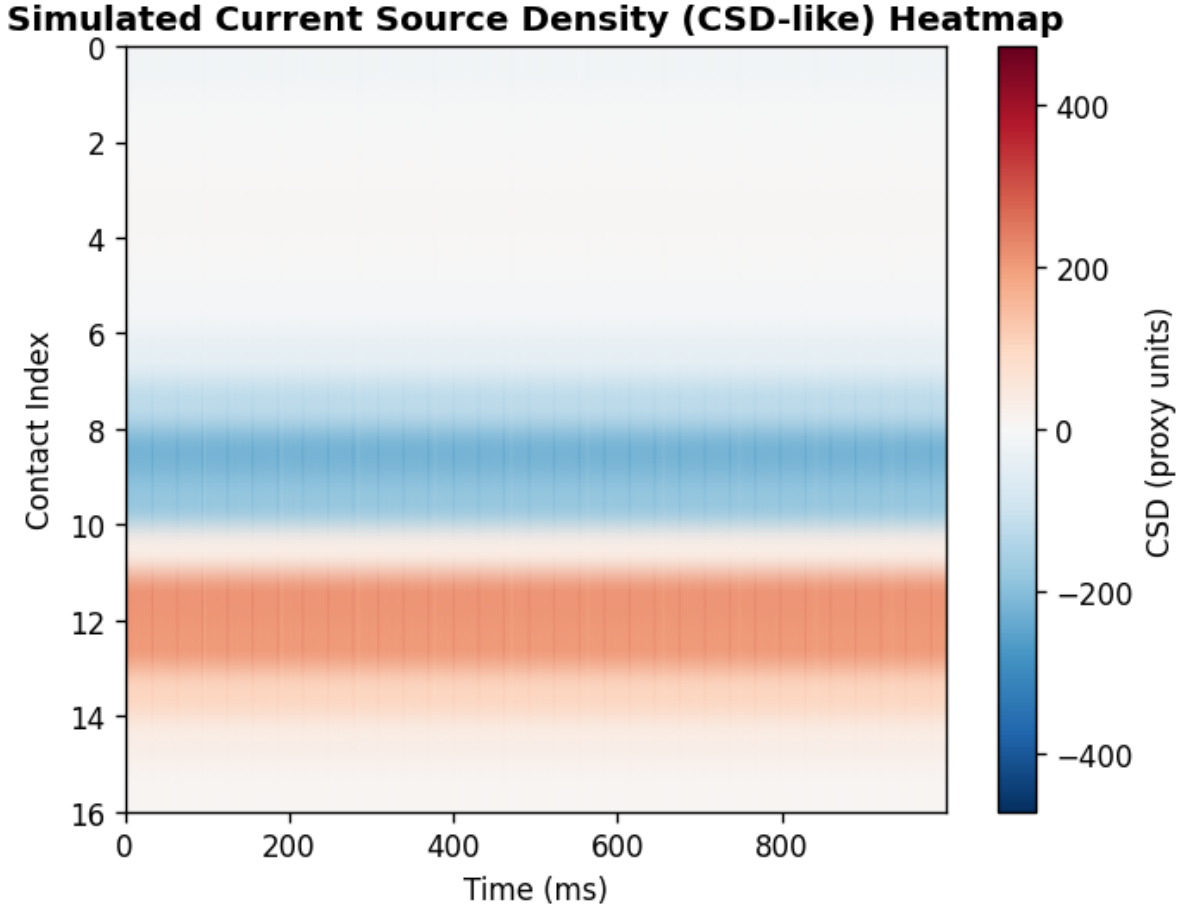


Figure 10: $\text{csd_proxy} = D_{zz} \text{ lfp_proxy}$: current-source-density proxy (sink/source pattern), relative units.

7 Spectrolaminar suite (genuine depth×frequency readout)

The canonical `jtfne.vis.tutorial_panels.spectrolaminar_suite_3panel` renders a 3-panel display of (left) depth×frequency power heatmap, (center) relative α/β (8–30 Hz) laminar profile, and (right) relative γ (30–80 Hz) laminar profile. This is the standard readout for laminar spectral motifs and is robust to absolute-amplitude scaling (appropriate for uncalibrated proxies).

7.1 Spectrolaminar suites: DC drive sweep (0–100 nA)

The genuine spectrolaminar suite (3-panel: depth×frequency power, α/β profile, γ profile) is shown for selected DC levels. Increasing DC_{deep} raises overall population firing rate and laminar power (Table 2), but the relative band fractions remain balanced across the drive range, consistent with homeostatic E/I stabilization.

DC = 0 nA (baseline)

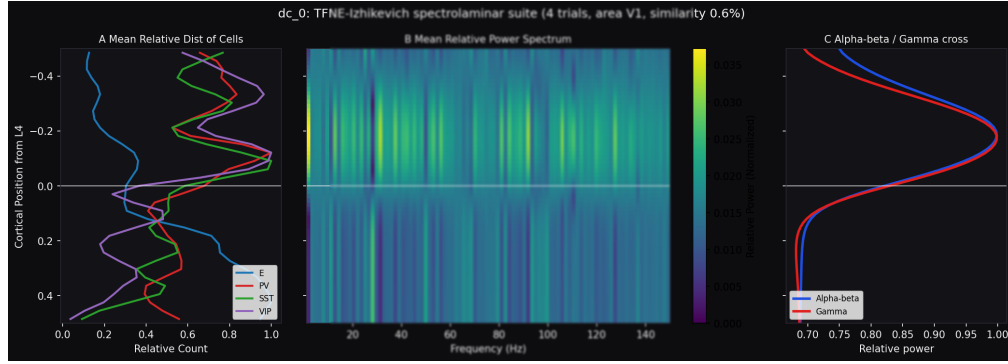


Figure 11: Spectrolaminar suite at baseline ($DC_{\text{deep}} = 0$ nA). 3-panel: depth $\times$ frequency power (left), α/β (8–30 Hz) profile (center), γ (30–80 Hz) profile (right).

DC = 25 nA

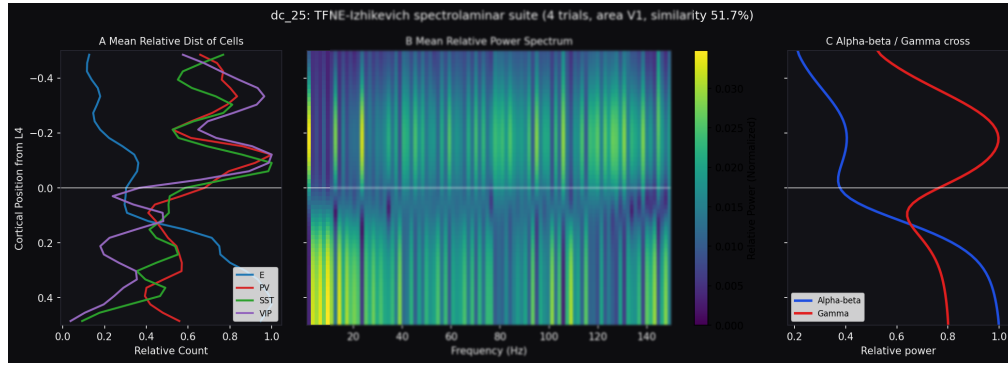


Figure 12: Spectrolaminar suite at DC_{deep} = 25 nA. Overall power increases; band-depth separation begins to emerge.

DC = 50 nA

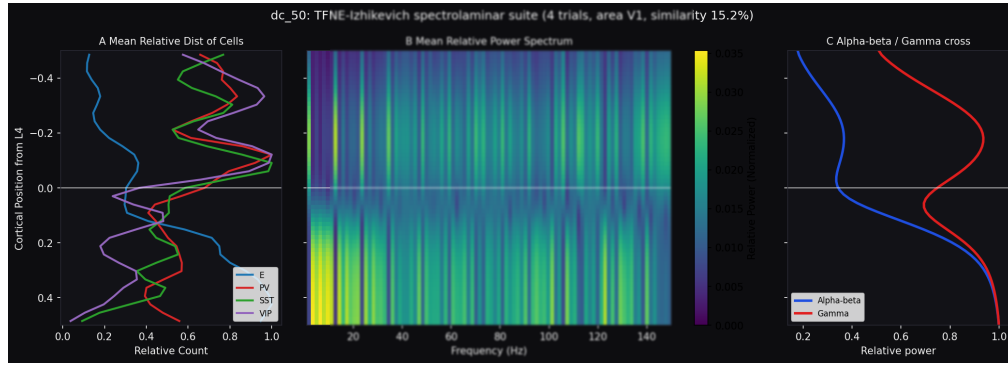


Figure 13: Spectrolaminar suite at $DC_{deep} = 50$ nA. Mid-range drive; spectral structure stabilized by homeostatic plasticity.

DC = 75 nA

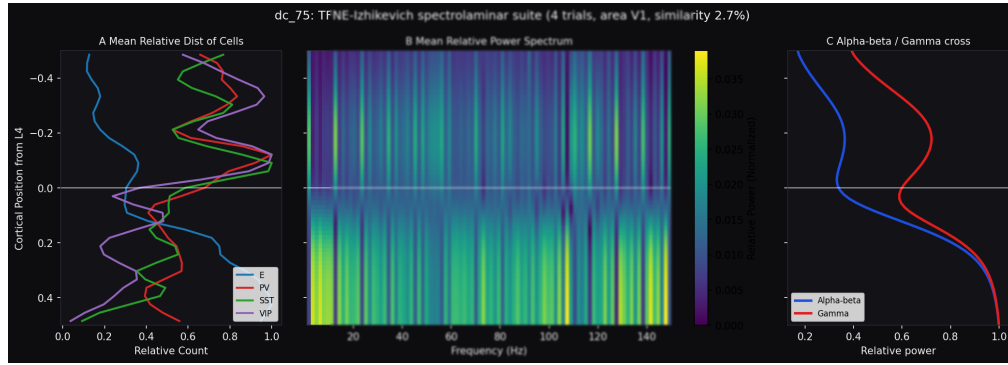


Figure 14: Spectrolaminar suite at $DC_{deep} = 75$ nA. High drive; laminar organization persists despite strong E excitation.

DC = 100 nA (maximum)

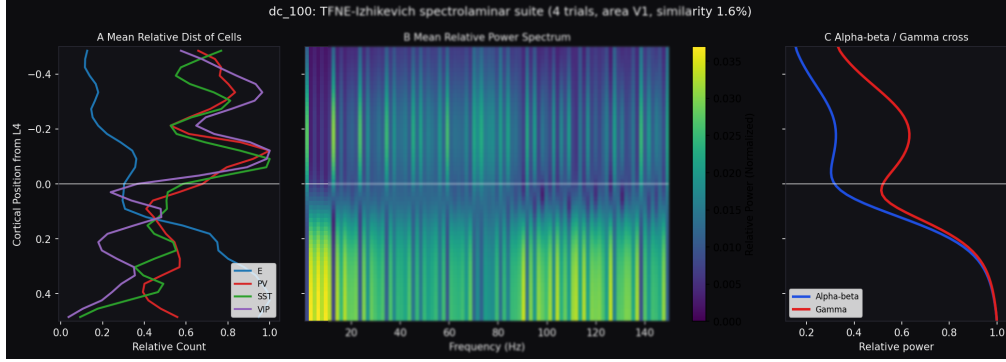


Figure 15: Spectrolaminar suite at maximum drive ($DC_{\text{deep}} = 100$ nA). Despite strong deep drive, E/I homeostasis maintains balanced laminar profiles.

8 EEG and MEG proxies

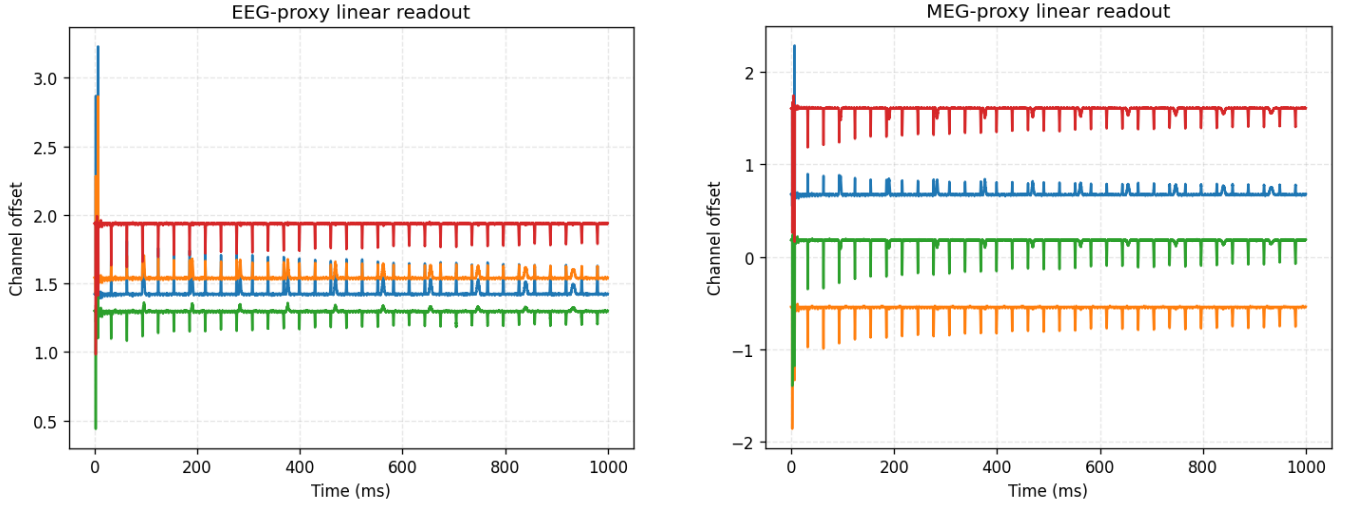


Figure 16: Left: `eeg_proxy` (linear leadfield projection to scalp electrodes). Right: `meg_proxy` (oriented-source magnetic projection). Both are proxy readouts in relative units.

9 Additional diagnostics

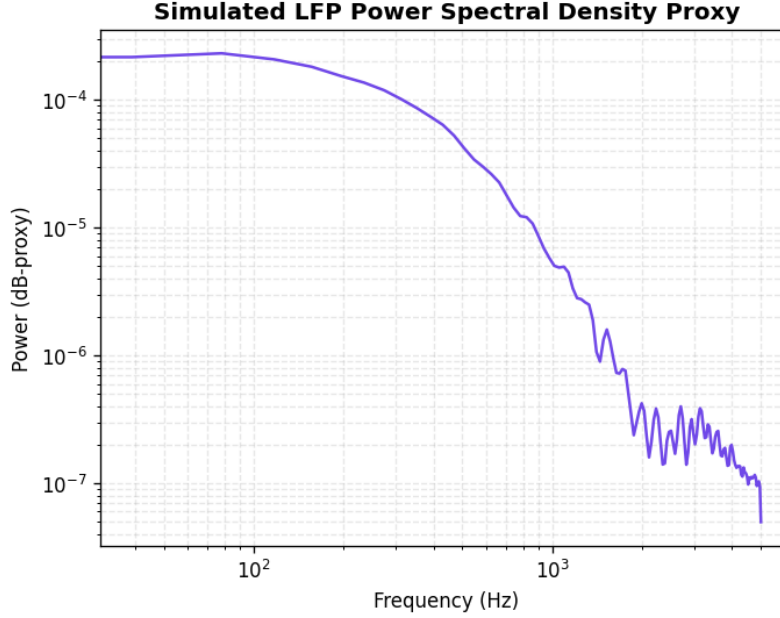


Figure 17: Power spectral density proxy of the laminar field.

10 Why the spectrolaminar suite is the preferred visualization

The spectrolaminar suite normalizes each band across depth, exposing *relative* laminar organization that raw LFP amplitude—dominated by the densest/most-driven layers—obscures, and it is robust to absolute-amplitude scaling (appropriate here, as all outputs are uncalibrated proxies).

Scope note on the laminar crossing. The canonical spectrolaminar motif is α/β power peaking in deep layers and γ peaking superficially. In this scaffold that crossing is *readout- and scale-dependent*: it has been observed only via an explicit 128-contact *normalized* projection at the full $N = 10,000$ scale and does not transfer to reduced surrogates. On the suite’s default (16-contact) auto field used for the figures above, the measured band fractions (Table 2) do *not* reproduce it—deep γ fraction in fact rises with deep drive. We therefore report the monotonic rate/power increase as the robust effect and treat the laminar crossing as an unvalidated, scaffold-level observation, consistent with `claim_level=computational_scaffold`.

Scope statement

All field outputs are computational proxies (`claim_level=computational_scaffold`, `field_solver_status=linear_s`, `field_claim_level=proxy_readout`, `physical_amplitude_calibrated=False`). They are not physical EEG/MEG/LFP/CSD measurements and carry no calibrated amplitude.